

Shubham Bawner

+91-7410730191 | [LinkedIn](#) | [Portfolio](#) | [GitHub](#) | shubhambawner.official@gmail.com

Backend and distributed systems engineer at Texas Instruments, designing AI-native agentic systems and robust and scalable backend services. I own Spring Boot services for customer support systems handling 80K+ monthly requests and an agentic support routing system saving 7K+ hours annually.

EXPERIENCE

Texas Instruments | Software Engineer | Bengaluru

Jul'24 – Present

TI E2E Ticket Routing System (Python, FastAPI, Elasticsearch, LLM, Webhooks)

- Architected and implemented an event driven AI based classification and ticket routing system.
- This system processes 500+ daily technical support queries, reducing overall manual triage effort by **20+ hours/day**.
- Designed a **multi-stage agentic** classification engine using RAG-based option pruning & dynamic prompt orchestration.
- Engineered a benchmark-driven eval framework, leading 15% routing accuracy improvement via iterative refinement.

Enterprise Support Intelligence Engine (powering the upcoming [ti.com](#) support chatbot) (Python, LlamaIndex, Kafka, FastAPI)

- Designed the planner agent for orchestrating parallel invocation of 10+ sub-agents and response consolidation for context-enrichment of an agentic system designed for **answering 7K+ enterprise support requests/daily**.
- Implemented the LlamaIndex event-driven asynchronous execution layer for sub-agents optimising response time.
- Built FastAPI SSE streaming layer, enabling real-time progress delivery to clients across a multi-agent orchestration pipeline.

Unified Customer Support Platform (Java 21, Spring Boot, Oracle DB, Redis, RabbitMQ)

- Owned backend services for ticket lifecycle for the platform handling **80K+ new support tickets/month** across 3 ticketing systems, covering payment, order, and return workflows.
- Engineered async post-actions processing service with RabbitMQ across a multi-node cluster, gaining **idempotency and fault tolerance**.
- Designed lookup and autocomplete services with **Redis caching** for hot prefixes for optimal response times.
- Built secure file upload service with virus scanning, extension spoofing detection, and legacy system compatibility.
- Resolved **critical production issues** affecting creation and routing of 5%+ tickets as part of hypercare and hotfix cycles.

Texas Instruments | Software Developer Intern | Bengaluru

Jan'24 – Jun'24

Centralised Alert & Autoheal Dashboard (FastAPI, Angular, Splunk, Icinga, Newrelic)

- Developed FastAPI backend services for alert management, threshold configuration, and role-based access control.

IIITPrep (EdTech) | Freelance Software Developer | Remote

Aug'23 – Dec'23

- Delivered an **email automation agent** integrating payment, shipping, and LMS APIs for contextual customer data retrieval, cutting weekly support workload from 15 hrs to 5 hrs.
- Dockerized the continuous deployment of the agentic application on AWS EC2.

EDUCATION

National Institute of Technology, Raipur | Bachelors, Information Technology

CGPA: 8.20/10 | Jul'24

SKILLS

Languages & Frameworks	Java, Python, JavaScript, SQL, FastAPI, Spring Boot, LlamaIndex
Databases & Tools	Oracle SQL, PostgreSQL, MS SQL, MongoDB, Redis, RabbitMQ, Docker, Jenkins, Git, Elasticsearch
Others	Distributed Systems, Async Messaging, Microservices, LLM Integration, RAG, Vector Search, AWS

ACHIEVEMENTS & CERTIFICATIONS

- **Rising Star Award (Q1'25)**, Texas Instruments | **TIITC'26**, TI India Technical Conference, Track: ASIT | **2nd Runner Up**, Codeutsava 6.0 Hackathon, NIT Raipur
- **LeetCode top 10%** | **500+ DSA** (LC, GFG, CodeChef, CF) | **JEE Mains'20**: Top 2% | **JEE Advanced'20**: Top 11K nationwide

[TIITC'26 Cert](#) | [NITRR Degree Cert](#) | [Deepthought Cert](#) | [AI Kalpa Cert](#)